NAMITHA M S

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Computer Science graduate specializing in Artificial Intelligence and Data Science with hands-on experience in software development, machine learning, computer vision, NLP, REST API development, and deep learning. Proficient in Python, C/C++, PyTorch, TensorFlow, OpenCV, FastAPI, ReactJS, NodeJS, SQL, MongoDB, Git, and Linux. Experienced in designing, developing, testing, debugging, optimizing, and deploying AI-enabled software solutions with emphasis on performance, scalability, and reliable inference pipelines.

EXPERIENCE

Pixl.AI **May 2025 – May 2026**
AI Engineering Intern Infopark, Kochi

- Designed and developed a real-time computer vision software system for deepfake detection using a Swin Transformer and EfficientNet-B0 ensemble architecture.
- Developed image preprocessing, feature extraction, inference, and model evaluation pipelines using Python, PyTorch, and OpenCV.
- Performed performance profiling, benchmarking, regression testing, and model validation to improve prediction accuracy and inference efficiency.

NIT Calicut **May 2024 – July 2024**
Research Intern Kozhikode, Kerala

- Conducted research on computer vision and image processing algorithms for skin cancer detection using unsupervised saliency detection techniques.
- Implemented spectral cluster voting algorithms for salient object detection in dermoscopic images.
- Developed image preprocessing, segmentation, feature extraction, and evaluation pipelines to improve detection robustness and model performance.

Pix Dynamics **June 2023 – July 2023**
Student Intern Infopark, Kochi

- Developed autoencoder-based image denoising models for signature restoration and image enhancement.
- Built dataset preparation, image preprocessing, and augmentation pipelines for computer vision and machine learning tasks.
- Worked with Python, PyTorch, OpenCV, YOLOv5, and deep learning frameworks for model development and optimization.

EDUCATION

Cochin University of Science and Technology
Integrated MSc in Computer Science (AI and Data Science) 2021 – April 2026

- Cumulative GPA: 8.44/10
- Specialization in Artificial Intelligence and Data Science

St Joseph's CGHSS, Thrissur
Higher Secondary Education 2018 – 2020

- Kerala State Higher Secondary Examination – 94%

TECHNICAL SKILLS

- **Programming:** Python, C, C++, R, SQL, JavaScript
- **Software Engineering:** DSA,REST APIs, FastAPI, Flask, NodeJS, ReactJS, Git, GitHub, Linux
- **AI/ML:** Machine Learning, Deep Learning, NLP, Computer Vision, Image Processing, Feature Extraction, Model Optimization
- **ML Frameworks:** PyTorch, TensorFlow, Hugging Face Transformers, OpenCV, YOLOv5
- **Databases:** SQL, MongoDB
- **Web Technologies:** HTML, CSS, JavaScript, ReactJS, NodeJS
- **Development:** Docker, API Development, Software Testing, Model Validation

PROJECTS

Passport Information Extraction API – OCR + Fine-Tuned LLM

- Developed an end-to-end software pipeline converting passport images into structured JSON using PaddleOCR and a LoRA fine-tuned Qwen 2.5 LLM.
- Built image preprocessing, OCR post-processing, schema extraction, batch inference, and validation pipelines.

- Developed and deployed a FastAPI REST API with Swagger documentation and optimized GPU/CPU inference.
- **Technologies:** Python, FastAPI, REST API, PaddleOCR, OpenCV, Hugging Face, Qwen, LoRA, PyTorch

Deepfake Detection using HyperSwin + rPPG

- Designed a multimodal deepfake detection architecture combining spatial facial features from Swin Transformer and EfficientNet-B0 with physiological rPPG-based temporal features.
- Implemented MediaPipe-based face detection and bilateral cheek Region of Interest (ROI) extraction, followed by green-channel signal extraction to obtain remote photoplethysmography (rPPG) time-series signals.
- Modeled temporal physiological patterns using LSTM and Bidirectional LSTM (BiLSTM) networks to capture inconsistencies in facial blood-flow signals across video frames.
- Extracted statistical physiological features including Pearson Correlation Coefficient (PCC), Standard Deviation (STD), and Mean Absolute Deviation (MAD) from left and right cheek signals to measure signal synchronization and temporal stability.
- Fused spatial deep-learning features with physiological and temporal features for binary real/fake classification, improving robustness against facial manipulation artifacts.
- Evaluated cross-dataset generalization using FaceForensics++ for training and Celeb-DF for testing, achieving 74.5% validation accuracy and 65.5% cross-dataset accuracy with the LSTM-based approach.
- **Technologies:** Python, PyTorch, OpenCV, MediaPipe, Swin Transformer, EfficientNet-B0, LSTM, BiLSTM, rPPG

Fake News Detection in Dravidian Languages

- Fine-tuned IndicBERT, MuRIL, and multilingual BERT models for Malayalam fake news classification.
- Designed text preprocessing and tokenization pipelines and evaluated model performance using classification metrics.
- Achieved 87.53% test accuracy using MuRIL with AdamW optimization and CrossEntropyLoss.
- **Technologies:** Python, PyTorch, Hugging Face Transformers, NLP

EBUY – Full-Stack Online Marketplace

- Developed a full-stack marketplace application with authentication, product listings, and backend services.
- Implemented MongoDB-based data storage and NodeJS APIs for backend functionality.
- Developed the frontend using HTML, CSS, and JavaScript.
- **Technologies:** JavaScript, NodeJS, MongoDB, HTML, CSS

ACHIEVEMENTS

- First Prize Winner – Hackify, Mar Athanasius College of Engineering
- TEDxCUSAT – Website UI Designer

CERTIFICATIONS

- IBM SkillsBuild – Artificial Intelligence Fundamentals, 2024
- IBM SkillsBuild – Getting Started with Artificial Intelligence, 2024
- NPTEL (IIT Madras) – Research Methodology, 2025
- NPTEL (IIT Guwahati) – Applied Accelerated Artificial Intelligence, 2025